

SURYANSH KUMAR PATHAK

Generative AI Engineer

Khanpur, Sahatwar, Ballia, Uttar Pradesh 277211, India | +91 9451345547 | suryanshkrpathak@gmail.com

[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Computer Science Engineering student with hands-on experience in Machine Learning, Deep Learning, Generative AI, Large Language Models, Retrieval-Augmented Generation, and Natural Language Processing. Skilled in building AI applications using Python, LangChain, Hugging Face, vector databases, and FastAPI, and deploying them with Docker. Familiar with LLM fine-tuning using LoRA and quantized LoRA, along with model deployment and full AI solution development from start to finish. Strong problem-solving skills with a focus on learning new technologies and building practical, real-world projects.

EDUCATION

B.Tech, Computer Science & Engineering — CGPA: 7.00 / 10.00

2023 – 2027 (Expected)

Shri Ramswaroop Memorial University | Lucknow, Uttar Pradesh, India

Intermediate (Class XII), Science (PCM)

2022 – 2023

M.V.V.M. School | Ballia, Uttar Pradesh, India

High School (Class X)

2020 – 2021

C.B Inter College | Ballia, Uttar Pradesh, India

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI / ML & GenAI: LangChain, RAG, Scikit-learn, TensorFlow, Keras, Hugging Face, vector database (ChromaDB, FAISS), Ollama

Deployment & MLOps: Docker, FastAPI, Streamlit, Git, GitHub

PROJECTS

RAG Copilot — AI Document Question-Answering System

- Built a hybrid-retrieval pipeline combining BM25 sparse search with ChromaDB dense retrieval over BAAI/bge-m3 embeddings, re-ranked via cross-encoder (ms-marco-MiniLM-L-6-v2), and orchestrated with LangChain LCEL
- Migrated the system from a proprietary LLM API to open-weights Qwen3-8B (Hugging Face), and built a Ragas evaluation harness (16 ground-truth Q&A pairs) to measure retrieval precision and faithfulness
- Diagnosed and fixed a production memory leak and an embedding-dimension mismatch surfaced during the model migration; deployed on Render, tuning chunk size and top-k retrieval to fit free-tier limits

Tools: Python, LangChain, ChromaDB, Hugging Face, Streamlit, Ragas, Git

Customer Support RAG Assistant

- Built a customer-support question-answering assistant using a hybrid retrieval pipeline that combines BM25 sparse retrieval with ChromaDB vector search, followed by Cross-Encoder re-ranking for improved context selection.
- Implemented the application with LangChain LCEL and Hugging Face BAAI/bge-m3 embeddings, with Groq openai/gpt-oss-20b for response generation.
- Added Ragas-based evaluation and observability to track query intents, latency, retrieval/re-ranking time, and errors; the captured dashboard shows 16 queries, 6.19 s average total latency, 4.58 s average retrieval + re-ranking latency, and 2 recorded errors.

Tools: Python, LangChain (LCEL), ChromaDB, BM25 (rank_bm25), Cross-Encoder Reranker, Groq openai/gpt-oss-20b, Hugging Face BAAI/bge-m3 Embeddings, Ragas, Streamlit

AI Medical Assistant

- Built an LLM-powered assistant analyzing symptoms and generating preliminary condition predictions with recommendation support, grounded via a FastAPI backend, vector database, and RAG pipeline
- Containerized the application with Docker and deployed it as a scalable service

Tools: Python, Transformers, LangChain, FastAPI, Docker

KEY ACHIEVEMENTS

- Smart India Hackathon: selected among the top teams in the college-level round for an AI-driven solution to a real-world problem
- Solved 250+ LeetCode problems across data structures and algorithms
- Certified by L&T as a Data Science Specialist